class14: RNASeq Mini-Project

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Background

The data for todays mini-project comes from a knock-down study of an important HOX gene, HOXa1.

Data Import

```
metadata <- read.csv("GSE37704_metadata.csv")
countdata <- read.csv("GSE37704_featurecounts.csv", row.names = 1)[,-1]
to.keep <- rowSums(countdata) > 0
countdata <- countdata[to.keep,]
metadata
```

```
         id     condition
1 SRR493366 control_sirna
2 SRR493367 control_sirna
3 SRR493368 control_sirna
4 SRR493369      hoxa1_kd
5 SRR493370      hoxa1_kd
6 SRR493371      hoxa1_kd
```

Cleanup

```r
colnames(countdata) == metadata$id
```

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

```r
nrow(countdata)
```

```
[1] 15975
```

DESeq Analysis

Setting up the DESeq object

```r
ctrl.inds <- metadata$condition == "control_sirna"
ctrl.inds
```

```
[1]  TRUE  TRUE  TRUE FALSE FALSE FALSE
```

```r
trtd.inds <- metadata$condition == "hoxa1_kd"
trtd.inds
```

```
[1] FALSE FALSE FALSE  TRUE  TRUE  TRUE
```

```r
ctrl.count <-  countdata[,ctrl.inds]
trtd.count <- countdata[,trtd.inds]
head(ctrl.count)
```

```
                SRR493366 SRR493367 SRR493368
ENSG00000279457        23        28        29
ENSG00000187634       124       123       205
ENSG00000188976      1637      1831      2383
ENSG00000187961       120       153       180
ENSG00000187583        24        48        65
ENSG00000187642         4         9        16
```

```
head(trtd.count)
```

```
                SRR493369 SRR493370 SRR493371
ENSG00000279457        29        28        46
ENSG00000187634       207       212       258
ENSG00000188976      1226      1326      1504
ENSG00000187961       236       255       357
ENSG00000187583        44        48        64
ENSG00000187642        14        16        16
```

```
ctrl.means <- rowMeans((ctrl.count))
head(ctrl.means)
```

```
ENSG00000279457 ENSG00000187634 ENSG00000188976 ENSG00000187961 ENSG00000187583
      26.666667      150.666667     1950.333333      151.000000       45.666667
ENSG00000187642
       9.666667
```

```
trtd.means <-  rowMeans(trtd.count)
head(trtd.means)
```

```
ENSG00000279457 ENSG00000187634 ENSG00000188976 ENSG00000187961 ENSG00000187583
       34.33333       225.66667      1352.00000       282.66667        52.00000
ENSG00000187642
       15.33333
```

```
means.df <- data.frame(ctrl.means, trtd.means)
head(means.df)
```

```
                 ctrl.means trtd.means
ENSG00000279457   26.666667   34.33333
ENSG00000187634  150.666667  225.66667
```

```
ENSG00000188976 1950.333333 1352.00000
ENSG00000187961  151.000000  282.66667
ENSG00000187583   45.666667   52.00000
ENSG00000187642    9.666667   15.33333
```

Running DESeq

```r
library(DESeq2)
```

```
Warning: package 'DESeq2' was built under R version 4.5.2

Warning: package 'S4Vectors' was built under R version 4.5.2

Warning: package 'IRanges' was built under R version 4.5.2

Warning: package 'GenomicRanges' was built under R version 4.5.2

Warning: package 'Biobase' was built under R version 4.5.2
```

```r
dds <- DESeqDataSetFromMatrix(countdata,
                              metadata,
                              ~condition)
```

```
Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
design formula are characters, converting to factors
```

```r
dds <- DESeq(dds)
dds
```

```
class: DESeqDataSet
dim: 15975 6
metadata(1): version
assays(4): counts mu H cooks
rownames(15975): ENSG00000279457 ENSG00000187634 ... ENSG00000276345
  ENSG00000271254
rowData names(22): baseMean baseVar ... deviance maxCooks
colnames(6): SRR493366 SRR493367 ... SRR493370 SRR493371
colData names(3): id condition sizeFactor
```

Getting results

```
res <- results(dds)
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)       : 4349, 27%
LFC < 0 (down)     : 4396, 28%
outliers [1]       : 0, 0%
low counts [2]     : 1237, 7.7%
(mean count < 0)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

Volcano Plot

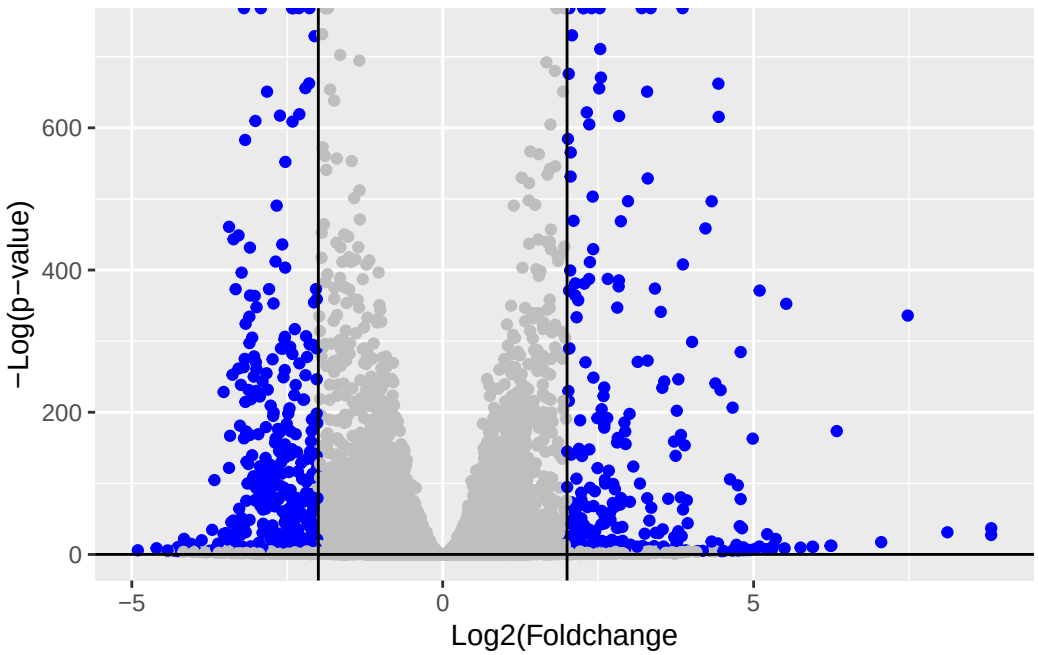
A plot of log2fold change vs -log(adjusted P-Value)

```
foldchanges <-  res$log2FoldChange

library(ggplot2)

prevolcano <- ggplot(res) +
  aes(foldchanges,
      -log(res$padj)) +
  geom_point()
```

```
library(ggplot2)

mycols <- rep("gray", "blue", nrow(res))
mycols[abs(foldchanges) > 2] <- "blue"
mycols[res$padj > 0.01 ] <- "gray"

volcano <- ggplot(res) +
  aes(foldchanges,
      -log(res$padj)) +
  geom_point(col = mycols) +
  labs(x = "Log2(Foldchange", y = "-Log(p-value)") +
```

```
  geom_vline(xintercept = c(-2,2)) +
  geom_hline(yintercept = .1)


volcano
```

```
Warning: Removed 1237 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Add Annotation

Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation to our results by completing the code below.

```
library("AnnotationDbi")
library("org.Hs.eg.db")

columns(org.Hs.eg.db)
```

```
 [1] "ACCNUM"       "ALIAS"        "ENSEMBL"      "ENSEMBLPROT"  "ENSEMBLTRANS"
```

```
 [6] "ENTREZID"     "ENZYME"       "EVIDENCE"     "EVIDENCEALL"  "GENENAME"
[11] "GENETYPE"     "GO"           "GOALL"        "IPI"          "MAP"
[16] "OMIM"         "ONTOLOGY"     "ONTOLOGYALL"  "PATH"         "PFAM"
[21] "PMID"         "PROSITE"      "REFSEQ"       "SYMBOL"       "UCSCKG"
[26] "UNIPROT"
```

```r
res$SYMBOL <- mapIds(org.Hs.eg.db,
                     keys = rownames(res),
                     keytype = "ENSEMBL",
                     column = "SYMBOL")
res$ENTREZID <- mapIds(org.Hs.eg.db,
                     keys = rownames(res),
                     keytype = "ENSEMBL",
                     column = "ENTREZID")
res$GENENAME <- mapIds(org.Hs.eg.db,
                     keys = rownames(res),
                     keytype = "ENSEMBL",
                     column = "GENENAME")
keggres <- res[!is.na(res$ENTREZID), ]
head(res)
```

```
log2 fold change (MLE): condition hoxa1 kd vs control sirna
Wald test p-value: condition hoxa1 kd vs control sirna
DataFrame with 6 rows and 9 columns
                 baseMean log2FoldChange     lfcSE       stat      pvalue
                <numeric>      <numeric> <numeric>  <numeric>   <numeric>
ENSG00000279457   29.9136      0.1792571 0.3248216   0.551863 5.81042e-01
ENSG00000187634  183.2296      0.4264571 0.1402658   3.040350 2.36304e-03
ENSG00000188976 1651.1881     -0.6927205 0.0548465 -12.630158 1.43990e-36
ENSG00000187961  209.6379      0.7297556 0.1318599   5.534326 3.12428e-08
ENSG00000187583   47.2551      0.0405765 0.2718928   0.149237 8.81366e-01
ENSG00000187642   11.9798      0.5428105 0.5215598   1.040744 2.97994e-01
                      padj      SYMBOL    ENTREZID               GENENAME
                 <numeric> <character> <character>            <character>
ENSG00000279457 6.86555e-01          NA          NA                     NA
ENSG00000187634 5.15718e-03      SAMD11      148398 sterile alpha motif ..
ENSG00000188976 1.76549e-35       NOC2L       26155 NOC2 like nucleolar ..
ENSG00000187961 1.13413e-07      KLHL17      339451 kelch like family me..
ENSG00000187583 9.19031e-01     PLEKHN1       84069 pleckstrin homology ..
ENSG00000187642 4.03379e-01       PERM1       84808 PPARGC1 and ESRR ind..
```

```r
write.csv(res,
          file="results_annotated.csv")
```

Pathway Analysis

```r
library(gage)
```

```r
library(gageData)
library(pathview)
```

```
##############################################################################
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG
license agreement (details at http://www.kegg.jp/kegg/legal.html).
##############################################################################
```

```r
data(kegg.sets.hs)
data(sigmet.idx.hs)
kegg.sets.hs = kegg.sets.hs[sigmet.idx.hs]
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
```

```
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

```r
res <- res[order(res$pvalue),]
write.csv(res, file = "deseq_results.csv")
```

KEGG

```r
keggres <- res[!is.na(res$ENTREZID), ]
foldchanges <- keggres$log2FoldChange
names(foldchanges) <- keggres$ENTREZID
keep <- !is.na(foldchanges) & is.finite(foldchanges)
foldchanges <- foldchanges[keep]
```

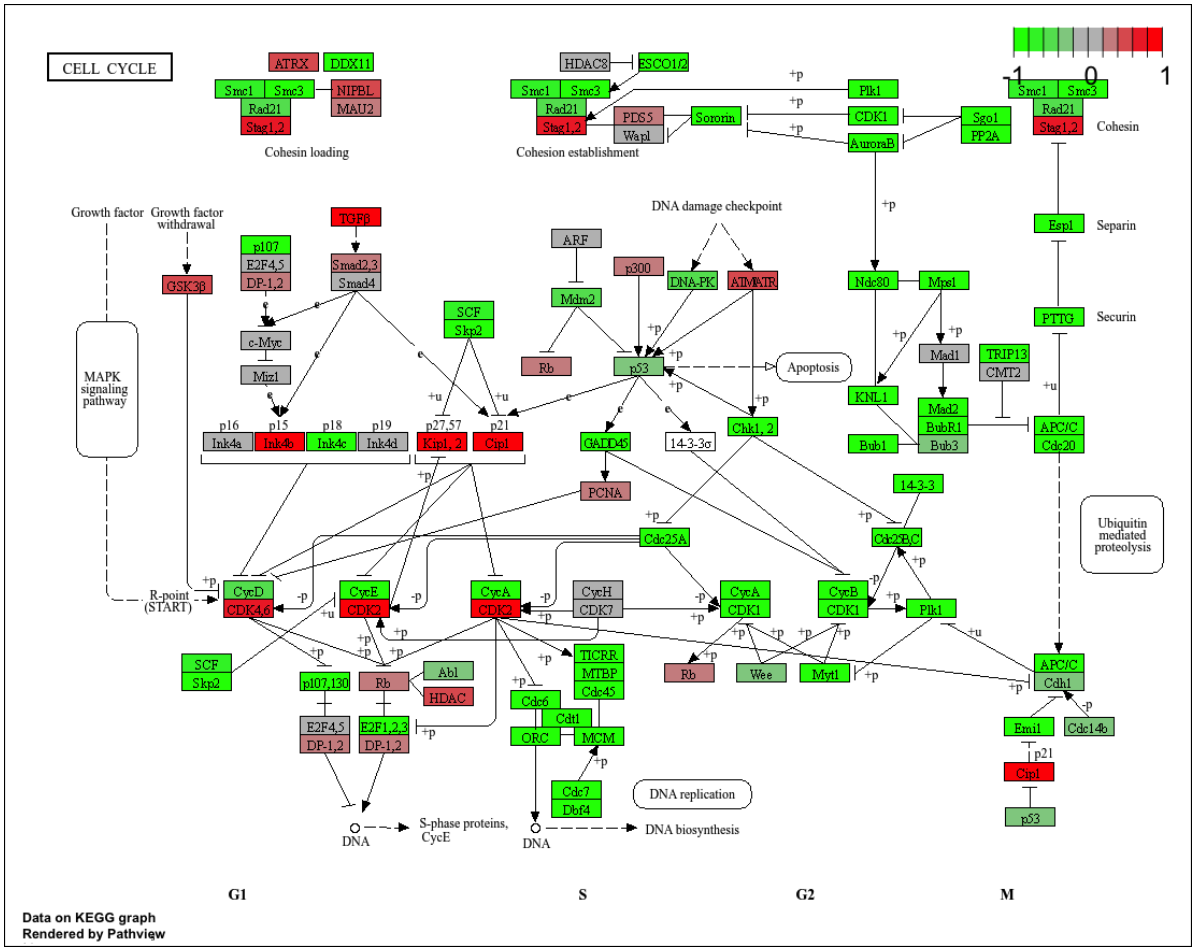
```r
library(pathview)
library(gage)
library(gageData)
```

```r
data(kegg.sets.hs)
data(sigmet.idx.hs)

kegg.sets.hs <- kegg.sets.hs[sigmet.idx.hs]
head(kegg.sets.hs, 3)
```

```
$`hsa00232 Caffeine metabolism`
[1] "10"   "1544" "1548" "1549" "1553" "7498" "9"

$`hsa00983 Drug metabolism - other enzymes`
 [1] "10"     "1066"   "10720"  "10941"  "151531" "1548"   "1549"   "1551"
 [9] "1553"   "1576"   "1577"   "1806"   "1807"   "1890"   "221223" "2990"
[17] "3251"   "3614"   "3615"   "3704"   "51733"  "54490"  "54575"  "54576"
[25] "54577"  "54578"  "54579"  "54600"  "54657"  "54658"  "54659"  "54963"
[33] "574537" "64816"  "7083"   "7084"   "7172"   "7363"   "7364"   "7365"
[41] "7366"   "7367"   "7371"   "7372"   "7378"   "7498"   "79799"  "83549"
[49] "8824"   "8833"   "9"      "978"

$`hsa00230 Purine metabolism`
  [1] "100"    "10201"  "10606"  "10621"  "10622"  "10623"  "107"    "10714"
  [9] "108"    "10846"  "109"    "111"    "11128"  "11164"  "112"    "113"
 [17] "114"    "115"    "122481" "122622" "124583" "132"    "158"    "159"
 [25] "1633"   "171568" "1716"   "196883" "203"    "204"    "205"    "221823"
 [33] "2272"   "22978"  "23649"  "246721" "25885"  "2618"   "26289"  "270"
 [41] "271"    "27115"  "272"    "2766"   "2977"   "2982"   "2983"   "2984"
 [49] "2986"   "2987"   "29922"  "3000"   "30833"  "30834"  "318"    "3251"
 [57] "353"    "3614"   "3615"   "3704"   "377841" "471"    "4830"   "4831"
 [65] "4832"   "4833"   "4860"   "4881"   "4882"   "4907"   "50484"  "50940"
 [73] "51082"  "51251"  "51292"  "5136"   "5137"   "5138"   "5139"   "5140"
 [81] "5141"   "5142"   "5143"   "5144"   "5145"   "5146"   "5147"   "5148"
 [89] "5149"   "5150"   "5151"   "5152"   "5153"   "5158"   "5167"   "5169"
 [97] "51728"  "5198"   "5236"   "5313"   "5315"   "53343"  "54107"  "5422"
[105] "5424"   "5425"   "5426"   "5427"   "5430"   "5431"   "5432"   "5433"
[113] "5434"   "5435"   "5436"   "5437"   "5438"   "5439"   "5440"   "5441"
[121] "5471"   "548644" "55276"  "5557"   "5558"   "55703"  "55811"  "55821"
[129] "5631"   "5634"   "56655"  "56953"  "56985"  "57804"  "58497"  "6240"
[137] "6241"   "64425"  "646625" "654364" "661"    "7498"   "8382"   "84172"
[145] "84265"  "84284"  "84618"  "8622"   "8654"   "87178"  "8833"   "9060"
[153] "9061"   "93034"  "953"    "9533"   "954"    "955"    "956"    "957"
[161] "9583"   "9615"
```

```r
pathview(gene.data=foldchanges, pathway.id="hsa04110")
```

```
'select()' returned 1:1 mapping between keys and columns

Info: Working in directory /Users/aidenquisenberry/Desktop/Biology Series/Bioinformatics/Rstu

Info: Writing image file hsa04110.pathview.png
```




```r
keggrespathways <- rownames(keggres$greater)[1:5]

keggresids = substr(keggrespathways, start=1, stop=8)
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
Info: Downloading xml files for hsaNA, 1/1 pathways..

Warning in download.file(xml.url, xml.target, quiet = T): cannot open URL
'https://rest.kegg.jp/get/hsaNA/kgml': HTTP status was '400 Bad Request'

Warning: Download of hsaNA xml file failed!
This pathway may not exist!

Info: Downloading png files for hsaNA, 1/1 pathways..

Warning: Download of hsaNA png file failed!
This pathway may not exist!

Warning: Failed to download KEGG xml/png files, hsa skipped!
```

```r
keggrespathways <- rownames(keggres$less)[1:5]

keggresids = substr(keggrespathways, start=1, stop=8)
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

```
Info: Downloading xml files for hsaNA, 1/1 pathways..

Warning in download.file(xml.url, xml.target, quiet = T): cannot open URL
'https://rest.kegg.jp/get/hsaNA/kgml': HTTP status was '400 Bad Request'

Warning: Download of hsaNA xml file failed!
This pathway may not exist!

Info: Downloading png files for hsaNA, 1/1 pathways..

Warning: Download of hsaNA png file failed!
This pathway may not exist!

Warning: Failed to download KEGG xml/png files, hsa skipped!
```

GO

```r
data(go.sets.hs)
data(go.subs.hs)

gobpsets = go.sets.hs[go.subs.hs$BP] # Focuses on Biological Process subset of GO

gobpres = gage(foldchanges, gsets=gobpsets)
lapply(gobpres, head)
```

```
$greater
                                            p.geomean stat.mean        p.val
GO:0007156 homophilic cell adhesion       8.132355e-05  3.836423 8.132355e-05
GO:0002009 morphogenesis of an epithelium 1.301491e-04  3.672346 1.301491e-04
GO:0048729 tissue morphogenesis           1.324453e-04  3.663702 1.324453e-04
GO:0007610 behavior                       1.786592e-04  3.585306 1.786592e-04
GO:0035295 tube development               5.558494e-04  3.273391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  5.613074e-04  3.277405 5.613074e-04
                                              q.val set.size         exp1
GO:0007156 homophilic cell adhesion       0.1804788      113 8.132355e-05
GO:0002009 morphogenesis of an epithelium 0.1804788      339 1.301491e-04
GO:0048729 tissue morphogenesis           0.1804788      424 1.324453e-04
GO:0007610 behavior                       0.1825897      426 1.786592e-04
GO:0035295 tube development               0.3325708      391 5.558494e-04
GO:0060562 epithelial tube morphogenesis  0.3325708      257 5.613074e-04

$less
                                              p.geomean stat.mean        p.val
GO:0048285 organelle fission             1.735928e-15 -8.047667 1.735928e-15
GO:0000280 nuclear division              4.808128e-15 -7.923688 4.808128e-15
GO:0007067 mitosis                       4.808128e-15 -7.923688 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.312717e-14 -7.781713 1.312717e-14
GO:0007059 chromosome segregation        2.149699e-11 -6.868718 2.149699e-11
GO:0000236 mitotic prometaphase          1.794819e-10 -6.689146 1.794819e-10
                                                 q.val set.size         exp1
GO:0048285 organelle fission             6.551876e-12      376 1.735928e-15
GO:0000280 nuclear division              6.551876e-12      352 4.808128e-15
GO:0007067 mitosis                       6.551876e-12      352 4.808128e-15
GO:0000087 M phase of mitotic cell cycle 1.341596e-11      362 1.312717e-14
GO:0007059 chromosome segregation        1.757594e-08      142 2.149699e-11
GO:0000236 mitotic prometaphase          1.222870e-07       84 1.794819e-10

$stats
```

```
                                          stat.mean     exp1
GO:0007156 homophilic cell adhesion        3.836423 3.836423
GO:0002009 morphogenesis of an epithelium  3.672346 3.672346
GO:0048729 tissue morphogenesis            3.663702 3.663702
GO:0007610 behavior                        3.585306 3.585306
GO:0035295 tube development                3.273391 3.273391
GO:0060562 epithelial tube morphogenesis   3.277405 3.277405
```

Reactome

```r
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "SYMBOL"]
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```r
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant "Entities p-value"? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

Cell Cycle, Mitotic has the most significant p-value at 2.1E-5. Yes, they both line up on the Cell Cycle being greatly affected by the treatment. The biggest difference would be from GO having a wider range of genes to pull from, and that their use in the cycle is not necassarily known. If GO found a major upregulation in a gene that doesnt really affect the cell cycle very greatly, it could provide a negative skew on how important that change in regulation really is.